2

MEASUREMENT AND SPACE

PYTHAGORAS' THEOREM

Pythagoras was an ancient Greek mathematician who lived in the 5th century BCE. The theorem (or rule) which carries his name was well known before his time, but Pythagoras may have been the first to prove it. Over 300 proofs for the theorem are known today. Pythagoras' theorem is the most famous mathematical formula, and it is still used today in architecture, engineering, surveying and astronomy.

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Chapter outline

	Working mathematically				
2.01 Surds	U	F		R	C
2.02 Pythagoras' theorem	U	F	PS	R	C
2.03 Finding the hypotenuse	U	F			
2.04 Finding a shorter side	U	F	PS	R	
2.05 Mixed problems	U	F	PS	R	
2.06 Testing for right-angled triangles	U	F		R	C
2.07 Pythagorean triads	U	F		R	C
2.08 Pythagoras' theorem problems	U	F	PS	R	

Wordbank

converse A rule or statement turned back-to-front; the reverse statement

hypotenuse The longest side of a right-angled triangle; the side opposite the right angle

irrational number A number such as π or $\sqrt{2}$ that cannot be expressed as a fraction

Pythagoras An ancient Greek mathematician who discovered an important formula about the sides of a right-angled triangle

Pythagorean triad A set of 3 positive integers that follow Pythagoras' theorem, such as {3, 4, 5}.

surd A square root (or other root) whose exact value cannot be found

theorem Another name for a formal rule or formula

Quiz
Wordbank 2

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (5):

2.01 Surds and irrational numbers
2.03, 2.04 Pythagoras' theorem 1 • Pythagoras' theorem 2
2.06 Testing for right-angled triangles
2.08 Pythagoras' theorem problems

***Twig* videos (3):**

2.02 Irrational numbers: Pythagoras • Proving Pythagoras • Building the pyramids

Quizzes (6):

- Wordbank 2
- SkillCheck 2
- Mental skills 2A
- Mental skills 2B
- Language of maths 2
- Test yourself 2

Skillsheet (1):

2.02 Pythagoras' theorem

Worksheets (8):

2.02 Pythagoras' discovery • A page of right-angled triangles • Pythagoras 1
2.04 Finding an unknown side
2.06 Testing for right-angled triangles
2.07 Pythagorean triads
2.08 Applications of Pythagoras' theorem
Mind map Pythagoras' theorem

Puzzles (4):

2.05 Pythagoras 1 • Pythagoras 2
2.08 Pythagorean two-step problems
Language of maths Pythagoras' theorem find-a-word

Technology (2):

2.03, 2.04 Pythagoras' theorem
2.07 Pythagorean triples

Spreadsheets (2):

2.03, 2.04 Pythagoras' theorem
2.07 Pythagorean triples

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- investigate irrational numbers and surds
- solve problems involving Pythagoras' theorem, writing the answers in decimal or surd form
- test whether a triangle is right-angled
- investigate Pythagorean triads

Quiz
SkillCheck 2

SkillCheck ANSWERS ON P. 565

1 Copy and complete:

a 50 cm = ____ mm **b** 200 cm = ____ m **c** 1.4 km = ____ m
d 62 mm = ____ cm **e** 8500 mm = ____ m **f** 1200 cm = ____ km

2 Evaluate each expression.

a 4^2 **b** 10.3^2 **c** $3^2 + 5^2$
d $8^2 - 6^2$ **e** $\sqrt{49}$ **f** $\sqrt{121}$

3 Round each number to 1 decimal place.

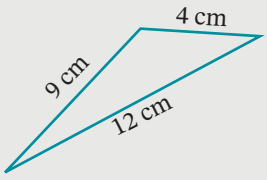
a 2.454 **b** 831.9342 **c** 372.658

4 Round each number to 2 decimal places.

a 28.777 **b** 7.9515 **c** 536.856

5 Find the perimeter of each shape.

a



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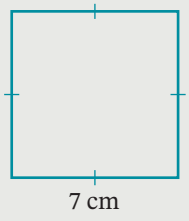

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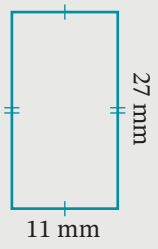
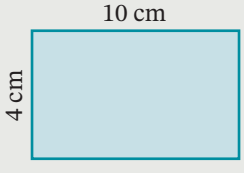


6 Find the area of each shape.

a



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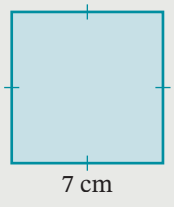

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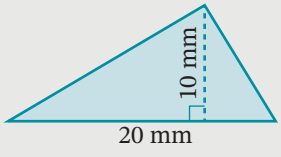

Surds

2.01

The **square** of a number is that number multiplied by itself. For example:

- $6^2 = 6 \times 6 = 36$ '6 squared' or 'the square of 6'
- $(-2)^2 = (-2) \times (-2) = 4$ '–2 squared' or 'the square of –2'

The **square root** ($\sqrt{\ }$) of a number is the positive value which, if squared, will give that number. For example:

- $\sqrt{25} = 5$ because $5^2 = 25$ 'the square root of 25'
- $\sqrt{81} = 9$ because $9^2 = 81$ 'the square root of 81'

Most square roots do not give exact answers like the ones above. For example, $\sqrt{7} = 2.645751311\ldots \approx 2.6$. Such roots are called **surds**.

A surd is a square root ($\sqrt{\ }$), cube root ($\sqrt[3]{\ }$), or any type of root whose exact decimal or fraction value cannot be found. As a decimal, its digits run endlessly *without repeating* (like π), so they are neither terminating nor recurring decimals.

Rational numbers, such as fractions, decimals and percentages, can be expressed in the form $\frac{a}{b}$, where a and b are integers (and $b \neq 0$), but surds are **irrational numbers** because they cannot be expressed in this form.

Example 1

Evaluate each expression.

a 5.3^2 **b** $(-18)^2$ **c** $7^2 + 9^2$

Solution

a $5.3^2 = 28.09$ On a calculator, enter: 5.3 x^2 =

b $(-18)^2 = 324$ (− 18) x^2 =

c $7^2 + 9^2 = 130$ 7 x^2 + 9 x^2 =

Example 2

Evaluate each expression, correct to two decimal places where necessary.

a $\sqrt{1024}$ **b** $\sqrt{89}$ **c** $\sqrt{15^2 + 19^2}$

Solution

a $\sqrt{1024} = 32$ √ 1024 =

b $\sqrt{89} = 9.433981...$
≈ 9.43 √ 89 =

c $\sqrt{15^2 + 19^2} = 24.207436$
≈ 24.21 √ (15 x^2 + 19 x^2) =

Video
Surds and irrational numbers

Example 3

Select the surds from this list of square roots: $\sqrt{72}$ $\sqrt{121}$ $\sqrt{64}$ $\sqrt{90}$ $\sqrt{28}$

Solution

$\sqrt{72} = 8.4852...$

$\sqrt{121} = 11$

$\sqrt{64} = 8$

$\sqrt{90} = 9.4868...$

$\sqrt{28} = 5.2915...$

so the surds are $\sqrt{72}$, $\sqrt{90}$ and $\sqrt{28}$.

$\sqrt{72}$ is called an '*exact* value' while 8.4852 is called its '*approximate* value'

EXERCISE 2.01 ANSWERS ON P. 565

Surds U F R C

EXAMPLE 1

1 Evaluate each expression.

a 4^2 **b** 16^2 **c** $(6.3)^2$ **d** $(0.7)^2$

e $(12.8)^2$ **f** $(0.02)^2$ **g** $(-1)^2$ **h** $(-9)^2$

i $(-13)^2$ **j** $(-1.2)^2$ **k** $2^2 + 7^2$ **l** $20^2 + 7^2$

m $19^2 - 14^2$ **n** $12^2 - 10^2$ **o** $(0.3)^2 + (0.4)^2$ **p** $(3.2)^2 - (1.5)^2$

2.01

EXAMPLE 2

2 Evaluate each square root.

a $\sqrt{196}$ **b** $\sqrt{900}$ **c** $\sqrt{625}$ **d** $\sqrt{361}$

e $\sqrt{1.69}$ **f** $\sqrt{2.89}$ **g** $\sqrt{10.24}$ **h** $\sqrt{156.25}$

3 Evaluate each square root, correct to 2 decimal places.

a $\sqrt{12}$ **b** $\sqrt{450}$ **c** $\sqrt{1001}$ **d** $\sqrt{325}$

e $\sqrt{153}$ **f** $\sqrt{7002}$ **g** $\sqrt{938}$ **h** $\sqrt{42}$

4 Which of the following is closest to the value of $\sqrt{6^2 + 11^2}$? Select **A**, **B**, **C** or **D**.

A 127 **B** 17 **C** 12.5 **D** 13.7

5 Evaluate each expression, correct to 2 decimal places.

a $\sqrt{5^2 + 7^2}$ **b** $\sqrt{6^2 + 11^2}$ **c** $\sqrt{17^2 - 12^2}$ **d** $\sqrt{8^2 - 3^2}$

6 Evaluate each expression, correct to one decimal place.

a $\sqrt{(1.5)^2 + (4.2)^2}$ **b** $\sqrt{(12.5)^2 - (7.1)^2}$ **c** $\sqrt{(25.7)^2 + (18.2)^2}$

EXAMPLE 3

7 Which of the following is a surd? Select **A**, **B**, **C** or **D**. C

A $\sqrt{9}$ **B** $\sqrt{225}$ **C** $\sqrt{160}$ **D** $\sqrt{81}$

8 Which of the following is NOT a surd? Select **A**, **B**, **C** or **D**. C

A $\sqrt{77}$ **B** $\sqrt{144}$ **C** $\sqrt{18}$ **D** $\sqrt{200}$

9 Select the surds from the following list of square roots. C

$\sqrt{32}$ $\sqrt{33}$ $\sqrt{289}$ $\sqrt{81}$ $\sqrt{4.9}$

$\sqrt{52}$ $\sqrt{121}$ $\sqrt{144}$ $\sqrt{196}$ $\sqrt{200}$

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10 Without using a calculator, determine between which 2 consecutive numbers the value of $\sqrt{40}$ lies. Select **A**, **B**, **C** or **D**. R

A 4 and 5 **B** 5 and 6 **C** 6 and 7 **D** 7 and 8

11 Is the statement $\sqrt{x^2} = x$ always true? R

12 Is the statement $\left(\sqrt{x}\right)^2 = x$ always true? R

Investigation

Pythagoras' theorem

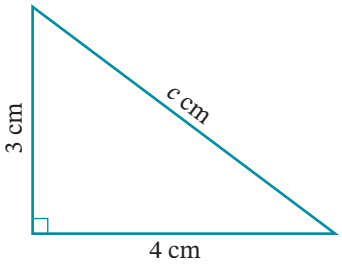
Pythagoras' theorem is a rule that describes the relationship between the longest side of a right-angled triangle and the other 2 (shorter) sides.

Use *dynamic geometry software* (or a pencil and ruler) to discover Pythagoras' theorem by following the steps below.

1 **a** Draw a right-angled triangle with the shorter sides of lengths 3 cm and 4 cm.

b Measure the length of the longest side, c cm.

c Copy the table below and enter the value for the longest side c when the 2 shorter sides are $a = 3$ and $b = 4$.



Shorter sides (cm)		Longest side (cm)				
a	b	c	a^2	b^2	$a^2 + b^2$	c^2
3	4					
6	2.5					
6	8					
5	12					

d Complete the table row by evaluating a^2, b^2, $a^2 + b^2$ and c^2.

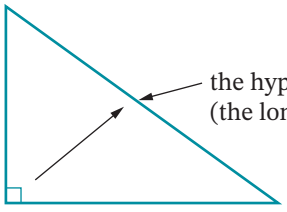
2 Complete the remaining rows by drawing right-angled triangles using the measurements given for the shorter sides a and b, finding the length of the longest side, c, and evaluating a^2, b^2, $a^2 + b^2$ and c^2.

3 What is the relationship between the longest side and the other 2 sides of a right-angled triangle?

Pythagoras' theorem

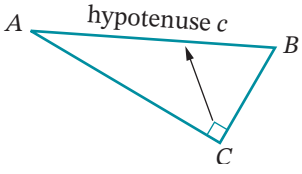
2.02

A **right-angled triangle** has one **right angle** (90°) and 2 acute angles. Its longest side is called the **hypotenuse**, which is always opposite the right angle.



The hypotenuse is always opposite the right angle.

The hypotenuse in this diagram is AB, opposite the right angle $\angle C$.

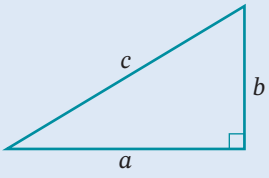


A **theorem** is a formal rule or formula. The ancient Greek mathematician Pythagoras discovered the following theorem.

Pythagoras' theorem

For any right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other 2 sides.

If c is the length of the hypotenuse, and a and b are the lengths of the other 2 sides, then:



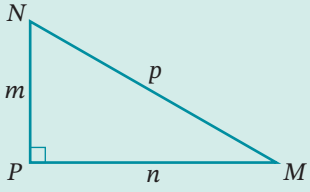
$$c^2 = a^2 + b^2$$

that is,

$$(\text{hypotenuse})^2 = (\text{shorter side 1})^2 + (\text{shorter side 2})^2$$

Example 4

Write **Pythagoras' theorem** for this triangle.



Solution

p is the hypotenuse, so $p^2 = m^2 + n^2$

OR

NM is the hypotenuse, so $NM^2 = NP^2 + PM^2$

Worksheets
Pythagoras' discovery
A page of right-angled triangles

Skillsheet
Pythagoras' theorem

Videos
Irrational numbers: Pythagoras
Proving Pythagoras
Building the pyramids

EXERCISE 2.02 ANSWERS ON P. 565

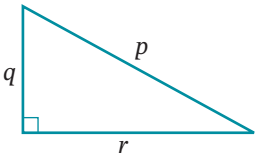
Pythagoras' theorem U F PS R C

1 In a right-angled triangle, what is the *hypotenuse*? Select the correct answer **A**, **B**, **C** or **D**. C

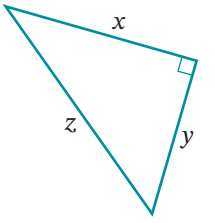
A the 90° angle **B** an acute angle **C** the shortest side **D** the longest side

2 For each right-angled triangle, name the hypotenuse. C

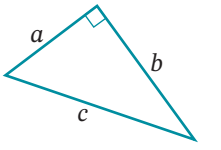
a



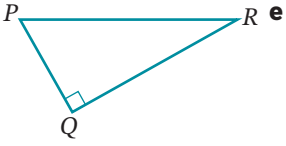
b



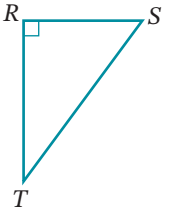
c



d



e



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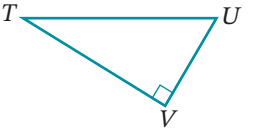
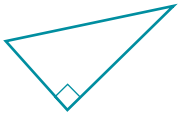


3 For each right-angled triangle, measure the length of the hypotenuse in millimetres (mm).

a



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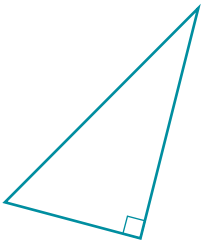

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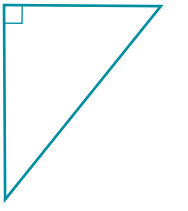

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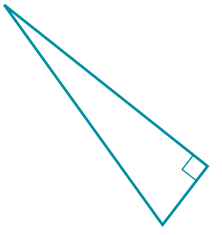

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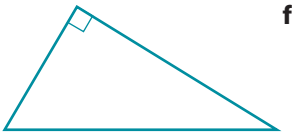

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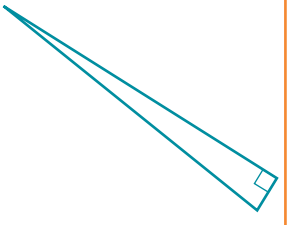

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Foundation Standard Complex

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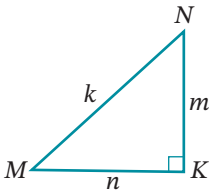
EXAMPLE 4

2.02

4 Write Pythagoras' theorem for each triangle. C

a



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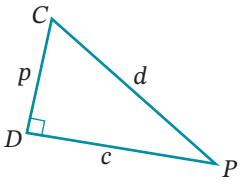

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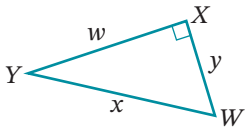

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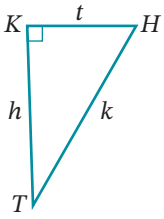

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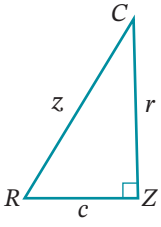

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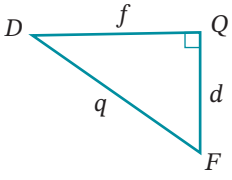
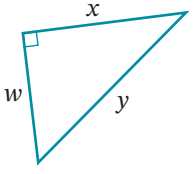


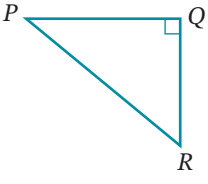
5 For each triangle, select the correct statement of Pythagoras' theorem. C

a



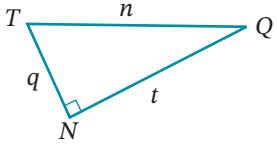
A $y^2 = x^2 + w^2$ **B** $x^2 = w^2 + y^2$
C $w^2 = x^2 + y^2$ **D** $y = x + w$

b



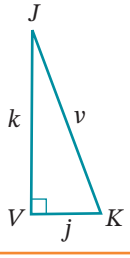
A $PR = PQ + QR$ **B** $QR^2 = PQ^2 + PR^2$
C $RP^2 = RQ^2 + PQ^2$ **D** $PQ^2 = PR^2 + QR^2$

c



A $q^2 = n^2 + t^2$ **B** $TN^2 + TQ^2 = NQ^2$
C $n^2 = q^2 + t^2$ **D** $TN^2 = TQ^2 + NQ^2$

d

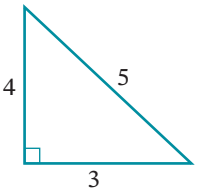


A $VK^2 + VJ^2 = JK^2$ **B** $j^2 = v^2 + k^2$
C $v^2 = j^2 + k^2$ **D** $VK^2 = JV^2 + JK^2$

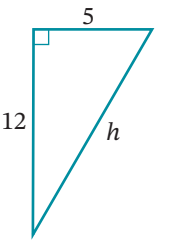
Foundation | Standard | Complex

6 Write Pythagoras' theorem for each triangle. C

a



b



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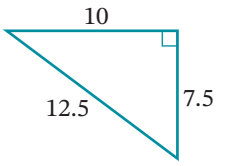

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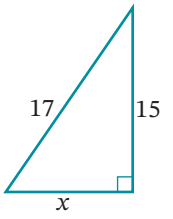
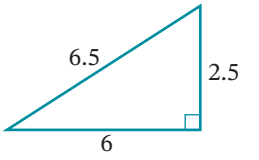


e



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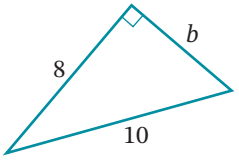
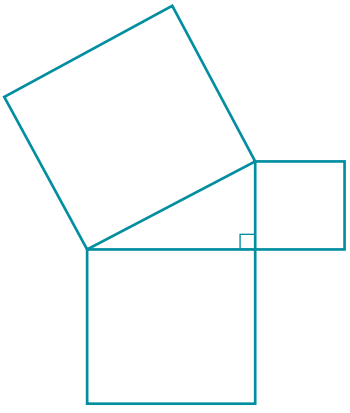


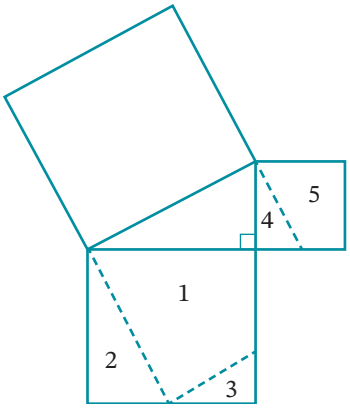
Worksheet
Pythagoras 1

7 Pythagoras' theorem can be demonstrated visually by dissection (cutting up shapes). PS R

a On a sheet of paper, construct a right-angled triangle and draw squares on all 3 sides. Alternatively, download the worksheet 'Pythagoras 1'.

b In the 2 smaller squares, draw lines perpendicular to the hypotenuse as shown, and in the bottom square, draw a line parallel to the hypotenuse as shown.





c Number the regions 1, 2, 3, 4 and 5 as shown.

d Cut out the 5 pieces and rearrange them to form the square on the hypotenuse.

e Does the area of the 2 smaller squares equal the area of the larger square?

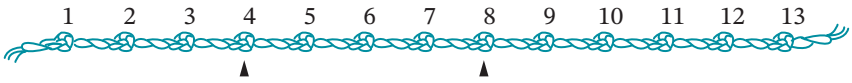
f How does this 'prove' Pythagoras' theorem?

g Repeat this question on another right-angled triangle.

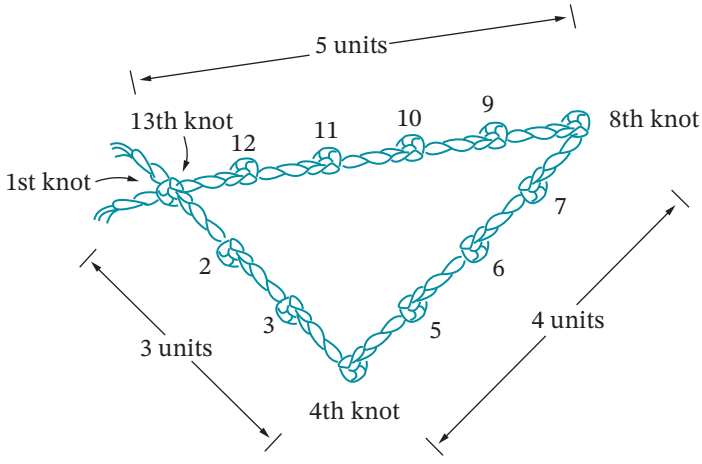
Did you know?

Egyptian rope stretchers

In ancient times, whenever the Nile River flooded, Egyptian farmers would have to rebuild fences around their fields afterwards. To form right angles for this, they used a secret 'magical' technique called rope stretching to create a 90° angle, making a length of rope with 13 knots tied in it at equal intervals.



The rope was fixed to the ground at the 4th and 8th knots, and stretched around to create a triangle with side lengths 3, 4 and 5 units. The right angle is formed at the 4th knot.



The builders of the great pyramids also used rope stretching, but could not understand why this 'trick' produced a right angle. Pythagoras studied with the rope stretchers and eventually worked it out.

Make and stretch a knotted rope to create a right angle using a 3-4-5 triangle.

Finding the hypotenuse 2.03

Pythagoras' theorem is used to find the length of one side of a right-angled triangle when the other 2 sides are known. In this section, we will look at finding the hypotenuse when the lengths of the 2 shorter sides are known.

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Technology
Pythagoras' theorem

Spreadsheet
Pythagoras' theorem

Video
Pythagoras' theorem 1

Example 5

Find the value of c in this triangle.

Solution

We want to find the length of the hypotenuse.

Using Pythagoras' theorem:

$c^2 = 9^2 + 40^2$

$= 1681$

$c = \sqrt{1681}$ Use the square root to find c.

$= 41$

An answer of $c = 41$ looks reasonable because:

- the hypotenuse is the longest side
- from the diagram, the hypotenuse looks a little longer than the side that is 40 cm

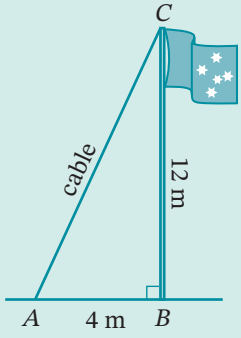
Video
Pythagoras' theorem 2

Example 6

Find the length of the cable supporting this flagpole:

a as a surd

b correct to one decimal place.



Solution

a $AC^2 = 12^2 + 4^2$

$= 160$

$AC = \sqrt{160}$ m This is the answer as a surd (in $\sqrt{\ }$ form)

b $AC = \sqrt{160}$ From part **a**

$= 12.6491...$

≈ 12.6 m Rounded to one decimal place

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Example 7

In $\triangle PQR$, $\angle P = 90°$, $PQ = 25$ cm and $PR = 32$ cm. Sketch the triangle and find the length of the hypotenuse, correct to one decimal place.

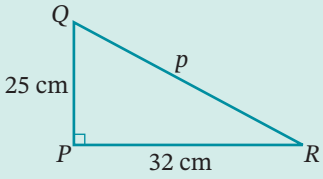
Solution

$p^2 = 32^2 + 25^2$
$= 1649$

p is the side opposite *P* and is also the hypotenuse

$p = \sqrt{1649}$
$= 40.607\,881\,01...$
≈ 40.6

$\therefore$ The length of the hypotenuse is 40.6 cm.

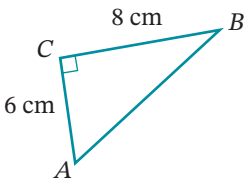


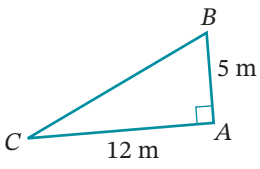
EXERCISE 2.03 ANSWERS ON P. 565

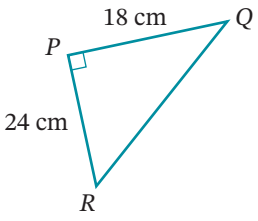
Finding the hypotenuse U F

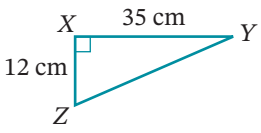
EXAMPLE 5

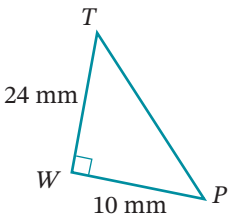
1 Find the length of the hypotenuse in each triangle.

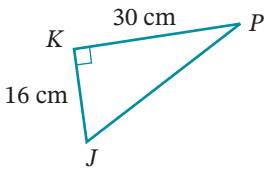
a


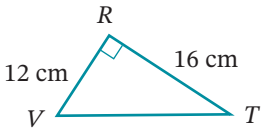
b


c


d


e


f


g


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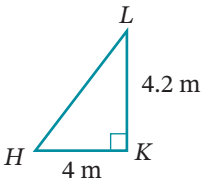

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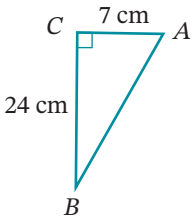

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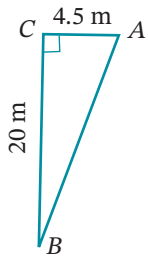
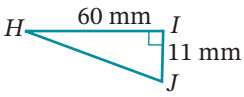


k


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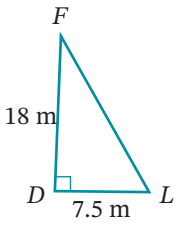
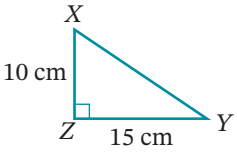


Foundation | Standard | Complex

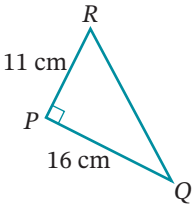
EXAMPLE 6

2 Find the length of the hypotenuse in each triangle, as a surd.

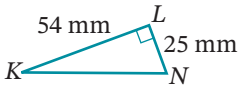
a



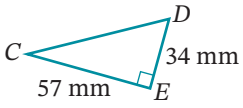
b



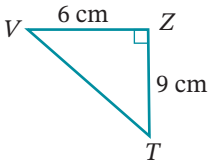
c



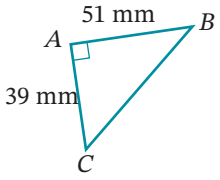
d



e

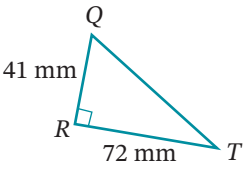


f

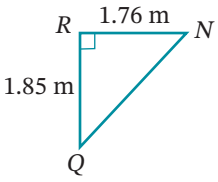


3 Find the length of the hypotenuse in each triangle, correct to one decimal place.

a



b



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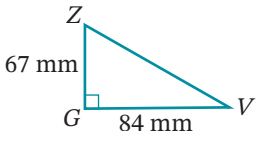

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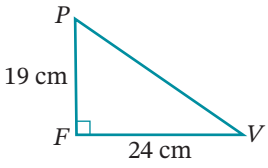

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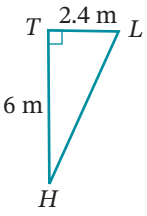

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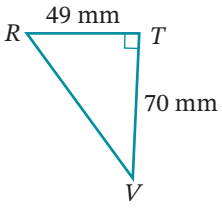
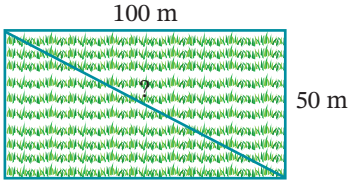


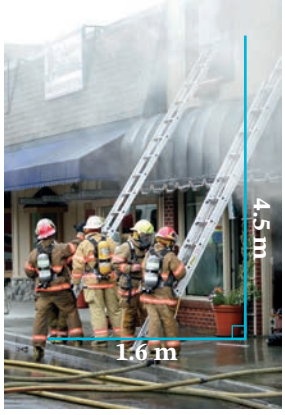
4 A rectangular field is 100 m long and 50 m wide. How far is it from one corner to the opposite corner, along the diagonal? Select the correct answer **A**, **B**, **C** or **D**.



A 150 m **B** 111.8 m

C 100.2 m **D** 98.3 m

5 A firefighter places a ladder on a window sill 4.5 m above the ground. If the foot of the ladder is 1.6 m from the wall, how long is the ladder, correct to one decimal place?



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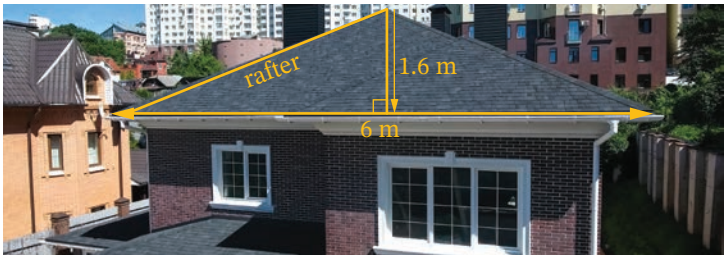
Foundation Standard Complex

9780170479370

6 The gable of a roof is 6 m long. If the roof is 1.6 m high, how long is the rafter?



Shutterstock.com/LuYago

EXAMPLE 7

7 **a** In $\triangle ABC$, $\angle ABC = 90°$, $AB = 39$ cm and $BC = 57$ cm. Find AC correct to one decimal place.

b In $\triangle MPQ$, $\angle PQM = 90°$, $QM = 2.4$ m and $PQ = 3.7$ m. Find PM, as a surd.

c In $\triangle RVJ$, $\angle J = 90°$, $JV = 12.7$ cm and $JR = 42$ mm. Find RV, correct to the nearest millimetre.

d In $\triangle EGB$, $EG = EB = 127$ mm and $\angle GEB = 90°$. Find the length of BG, correct to one decimal place.

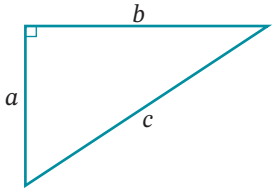
e In $\triangle VZX$, $\angle V = 90°$, $VX = 547$ cm and $VZ = 11.6$ m. Find ZX in metres, correct to the nearest 0.1 m.

f In $\triangle PQR$, $\angle RPQ = 90°$, $PQ = 2350$ mm and $PR = 5.8$ m. Find QR in metres, correct to 2 decimal places.

Technology

Finding the hypotenuse

Use a spreadsheet to calculate the length of the hypotenuse, c, of a right-angled triangle, given the lengths of the other two sides (a and b).



1 Enter the headings shown below into cells A1, B1, C1 and D1. Then enter the values shown into columns A and B.

	A	B	C	D	E
1	a	b	a^2+b^2	c	
2	3	4			
3	5	12			
4	6	8			
5	7	24			
6					
7					

2 Click on cell C2. Type the formula **=A2^2+B2^2**. Use **Fill Down** to copy this formula into cells C3 to C5.

3 To calculate each hypotenuse, we need to find the square root of the values in column C each time. So, in cell D2, enter **=sqrt(C2)**. Then click on cell D2 and use **Fill Down** to copy this formula into cells D3 to D5.

4 Use your spreadsheet to find the length of the hypotenuse, given the following pairs of values for the other two sides.

a $a = 12, b = 9$ **b** $a = 10, b = 10$ **c** $a = 7, b = 6$

d $a = 1, b = 8$ **e** $a = 7.5, b = 11.9$ **f** $a = 6.4, b = 2.3$

Foundation Standard Complex

2.04 Finding a shorter side

Pythagoras' theorem can also be used to find the length of a shorter side of a right-angled triangle, if the hypotenuse and the other side are known.

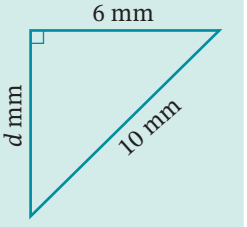
Worksheet
Finding an unknown side

Technology
Pythagoras' theorem

Spreadsheet
Pythagoras' theorem

Example 8

Find the value of d in this triangle.



Video
Pythagoras' theorem 2

Solution

We want to find the length of a shorter side.

Using Pythagoras' theorem:

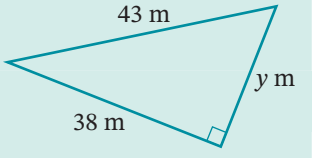
$10^2 = d^2 + 6^2$
$100 = d^2 + 36$
$d^2 + 36 = 100$
$d^2 = 100 - 36$
$= 64$
$d = \sqrt{64}$
$= 8$

From the diagram, a length of 8 mm looks reasonable because it must be shorter than the hypotenuse, which is 10 mm.

Video
Pythagoras' theorem 1

Example 9

Find the value of y as a surd for this triangle.



Solution

$43^2 = y^2 + 38^2$
$1849 = y^2 + 1444$
$y^2 + 1444 = 1849$
$y^2 = 1849 - 1444$
$= 405$
$y = \sqrt{405}$ Leave the answer as a surd

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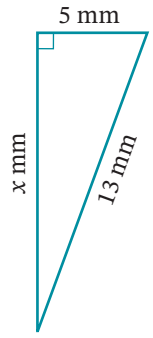
EXERCISE 2.04 ANSWERS ON P. 565

Finding a shorter side U F PS R

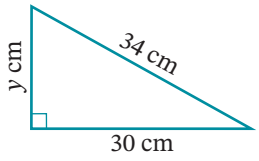
EXAMPLE 8

1 Find the value of the variable in each triangle.

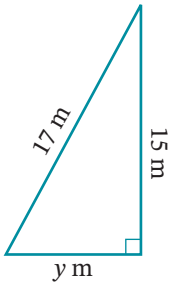
a



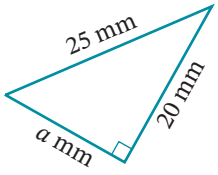
b



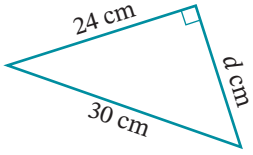
c



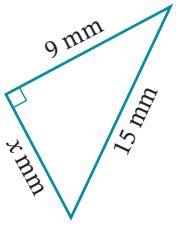
d



e

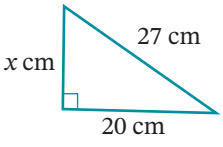


f



2 Find the value of the variable in each triangle, correct to one decimal place.

a



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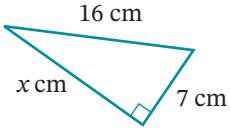

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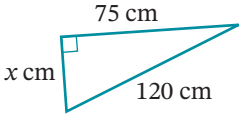
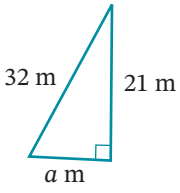


d



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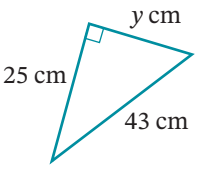

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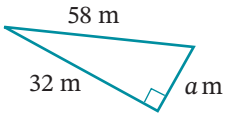
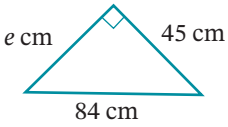


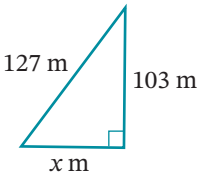
EXAMPLE 9

3 Find the value of the variable in each triangle as a surd.

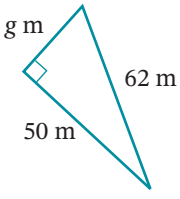
a



b



c



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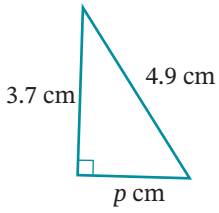
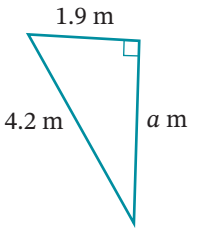
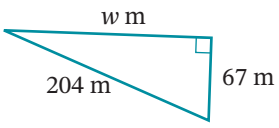
d



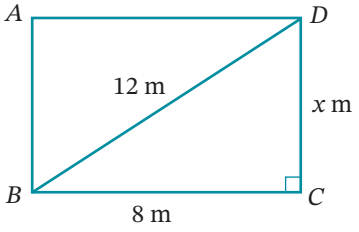
e



f



4 Find the value of x in this rectangle. Select the correct answer **A**, **B**, **C** or **D**.



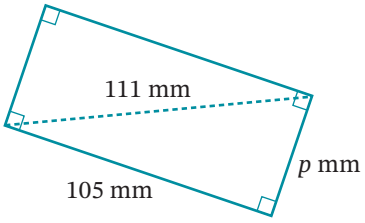
A $\sqrt{20}$

B $\sqrt{80}$

C $\sqrt{136}$

D $\sqrt{208}$

5 Find the value of p in this rectangle. Select **A**, **B**, **C** or **D**.



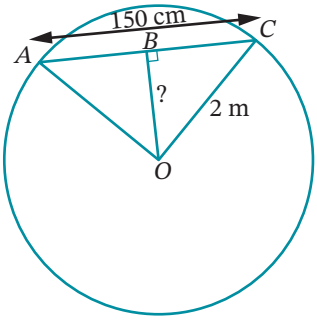
A 40

B 36

C 32

D 28

6 In this diagram, O is the centre of a circle.
A perpendicular line is drawn from O to B such that $OB \perp AC$ and $AB = BC$. Calculate the length of OB.
Select **A**, **B**, **C** or **D**. PS R



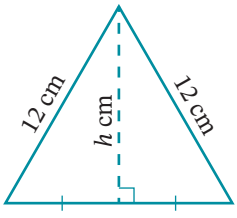
A 1.3 m

B 1.8 m

C 1.9 m

D 3.4 m

7 An equilateral triangle has sides of length 12 cm. Find the perpendicular height, h cm, of the triangle, correct to 2 decimal places. PS R



8 A square has a diagonal of length 30 cm. What is the length of each side of the square, correct to the nearest millimetre? PS R

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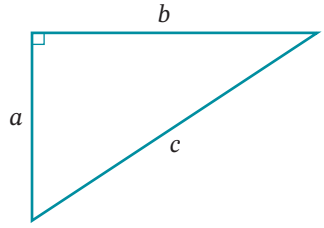
Technology

Finding a shorter side

Use a spreadsheet to calculate the length of a shorter side, b, of a right-angled triangle, given the length of the hypotenuse and the other side (c and a).



1 Enter the headings shown below into cells A1, B1, C1 and D1. Then enter the values shown into columns A and B.

	A	B	C	D
1	c	a	c^2 - a^2	b
2	25	20		
3	37	12		
4	18	16		
5	11	7		
6				

2 Click on cell C2. Type the formula **=A2^2-B2^2**. Use **Fill Down** to copy this formula into cells C3 to C5.

3 To calculate the values of b, we need to find the square root of each of the values in column C. So, in cell D2, enter **=sqrt(C2)**. Then click on cell D2 and use **Fill Down** to copy this formula into cells D3 to D5.

4 Use your spreadsheet to find the length of the unknown shorter side, given the following pairs of values for the hypotenuse and other side.

a $c = 61, a = 11$

b $c = 87, a = 60$

c $c = 35, a = 22$

d $c = 9.2, a = 2.7$

e $c = 158, a = 63$

f $c = 24.5, a = 12.8$

Quiz
Mental skills 2A

Mental skills 2A: Maths without calculators ANSWERS ON P. 565

Multiplying and dividing by a power of 10

Multiplying a number by 10, 100, 1000, etc. moves the decimal point to the right and makes the number bigger. We place zeros at the end of the number if necessary.

- When multiplying a number by **10**, move the decimal point **1** place to the right.
- When multiplying a number by **100**, move the decimal point **2** places to the right.
- When multiplying a number by **1000**, move the decimal point **3** places to the right.

The number of places the decimal is moved to the right matches the number of 0s in the 10, 100 or 1000 we are multiplying by.

1 Study each example.

a $26.32 \times 10 = 26.32 = 263.2$ The point moves 1 place to the right.

b $8.701 \times 100 = 8.701 = 870.1$ The point moves 2 places to the right.

c $6.01 \times 1000 = 6.010 = 6010$ The point moves 3 places to the right after a zero is placed at the end.

d $17 \times 100 = 17.00 = 1700$ The point moves 2 places to the right after 2 zeros are placed at the end.

2 Now evaluate each expression.

a	89.54×10	**b**	3.7×10	**c**	0.831×100
d	42×100	**e**	5.2716×1000	**f**	156.1×10
g	31.84×1000	**h**	64.3×100	**i**	0.0224×1000
j	4.894×10	**k**	7.389×1000	**l**	11.42×100

Dividing a number by 10, 100, 1000, etc. moves the decimal point to the left and makes the number smaller. We place 0s at the start of the decimal if necessary.

- When dividing a number by **10**, move the decimal point **1** place to the left.
- When dividing a number by **100**, move the decimal point **2** places to the left.
- When dividing a number by **1000**, move the decimal point **3** places to the left.

3 Study each example.

a $145.66 \div 10 = 145.66 = 14.566$ The point moves 1 place to the left.

b $2.357 \div 100 = 002.357 = 0.023\,57$ The point moves 2 places to the left after 2 zeros are inserted at the start.

c $14.9 \div 1000 = 0014.9 = 0.0149$ The point moves 3 places to the left after 2 zeros are inserted at the start.

d $45 \div 100 = 045. = 0.45$ The point moves 2 places to the left after 1 zero is inserted at the start.

4 Now evaluate each expression.

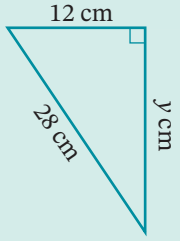
a	$733.4 \div 10$	**b**	$9.4 \div 10$	**c**	$652 \div 100$
d	$10.4 \div 100$	**e**	$704 \div 1000$	**f**	$198.5 \div 100$
g	$2 \div 100$	**h**	$4159 \div 1000$	**i**	$123 \div 10$
j	$0.758 \div 100$	**k**	$8.49 \div 100$	**l**	$25.1 \div 1000$

Mixed problems 2.05

In this section, you will need to decide whether to find the hypotenuse or a shorter side, using Pythagoras' theorem.

Example 10

Find the value of y as a surd.



Solution

y is the length of a shorter side.

$$28^2 = y^2 + 12^2$$
$$784 = y^2 + 144$$
$$y^2 + 144 = 784$$
$$y^2 = 784 - 144$$
$$= 640$$
$$y = \sqrt{640}$$

Example 11

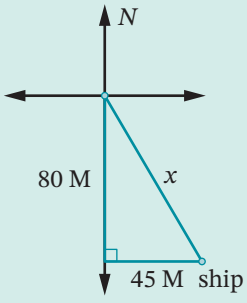
A ship sails 80 nautical miles south and then 45 nautical miles east. How far is it from its starting point, correct to one decimal place?

Solution

Let x be the distance the ship is from the starting point.

$$x^2 = 80^2 + 45^2$$
$$= 8425$$
$$x = \sqrt{8425}$$
$$= 91.7877...$$
$$\approx 91.8\text{M}$$

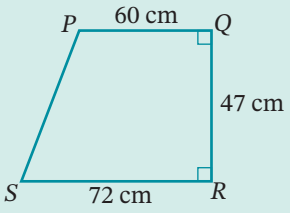
From the diagram, this looks like a reasonable answer



M is the symbol for nautical miles

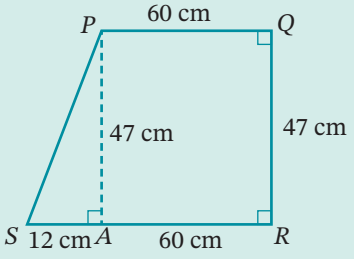
Example 12

Find the length of the unknown side in this trapezium, correct to one decimal place.



Solution

The unknown side is PS. Form the right-angled $\triangle PSA$, as shown.



By comparing sides:

$PA = QR = 47$ cm and
$AR = PQ = 60$ cm

$\therefore SA = 72 - 60 = 12$ cm

$\therefore PS^2 = 47^2 + 12^2$

$= 2353$

$PS = \sqrt{2353}$

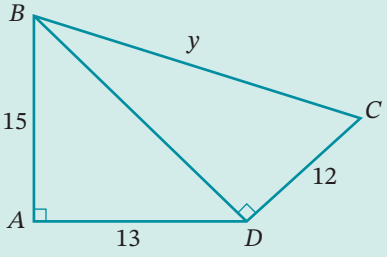
$= 48.5077...$

≈ 48.5 cm

From the diagram, this looks like a reasonable answer.

Example 13

Find the value of y, correct to 2 decimal places.



Solution

We need to find BD first.

In $\triangle ABD$,

$BD^2 = 15^2 + 13^2$

$= 394$

$BD = \sqrt{394}$ Leave BD as a surd for further working

In $\triangle BCD$,

$y^2 = \left(\sqrt{394}\right)^2 + 12^2$ $\left(\sqrt{394}\right)^2 = 394$

$= 538$

$y = \sqrt{538}$

$= 23.1948...$

≈ 23.19

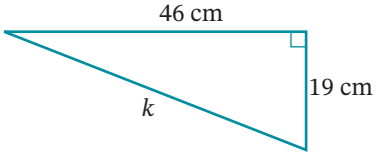
From the diagram, this looks like a reasonable answer.

EXERCISE 2.05 ANSWERS ON P. 566

Mixed problems U F PS R

EXAMPLE 10

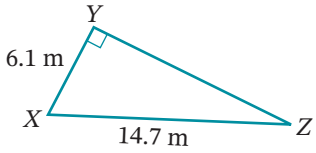
1 What is the length of k? Select the correct answer **A**, **B**, **C** or **D**.



A 27 cm **B** 41.9 cm

C 46.2 cm **D** 49.8 cm

2 What is the length of YZ? Select **A**, **B**, **C** or **D**.

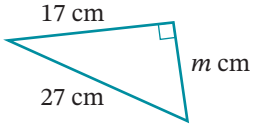


A 15.9 m **B** 13.4 m

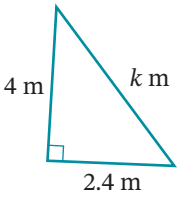
C 11.7 m **D** 8.6 m

3 Find the value of the variable in each triangle. Give your answers correct to one decimal place.

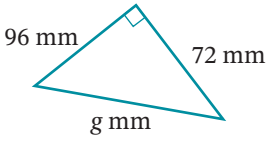
a



b



c



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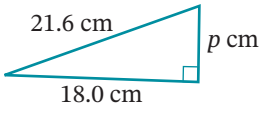
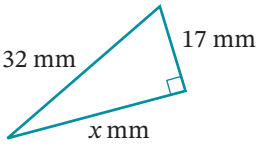
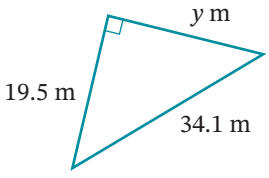


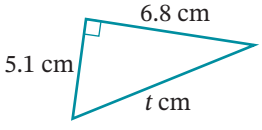
e



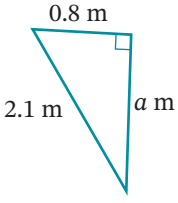
f



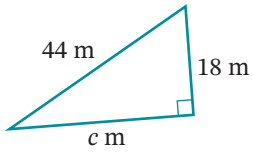
g



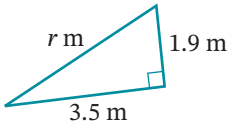
h



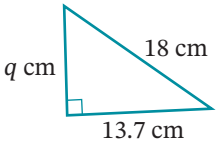
i



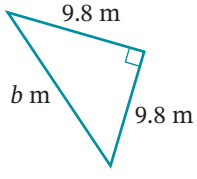
j



k



l



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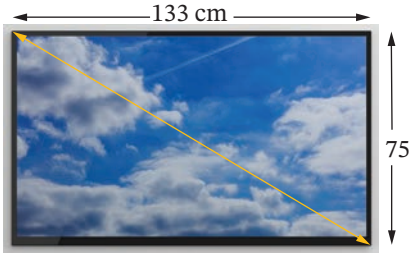
4 A ladder 5 m long is leaning against a wall. If the base of the ladder is 2 m from the bottom of the wall, how far does the ladder reach up the wall? (Answer correct to 2 decimal places.)

iStock.com/bmcent1

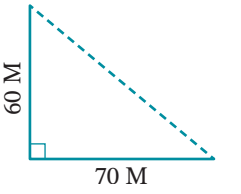
5 The size of a TV screen is described by the length of its diagonal. If a TV screen is 133 cm wide and 75 cm high, what is the size of the screen? Answer to the nearest centimetre.



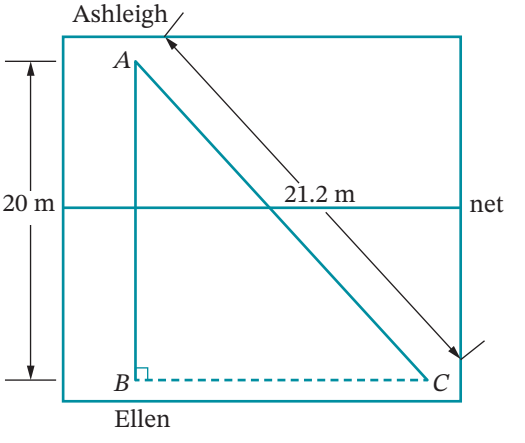
Shutterstock.com/rawf8

EXAMPLE 11

6 A ship sails 70 nautical miles west and then 60 nautical miles north. How far is it from its starting point, correct to one decimal place?



7 Ashleigh is standing on a tennis court at A and serves the ball to Ellen, who hits it back from B. Ashleigh then hits the ball from A to C. How far does Ellen have to run to hit a return shot? Answer to the nearest metre.



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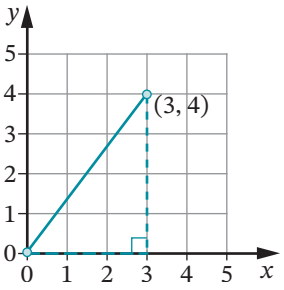
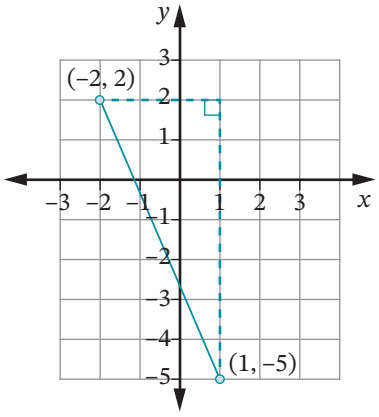
8 Use Pythagoras' theorem to calculate the distance between the 2 points on each number plane. In part **b**, write the answer as a surd. R

a



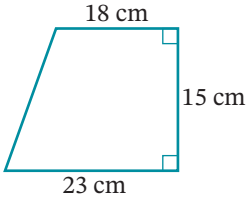
b



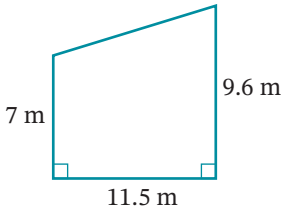
9 For each trapezium, find the length of the unknown side, correct to 2 decimal places where necessary. R

EXAMPLE 12

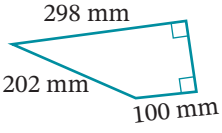
a



b

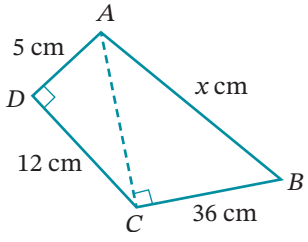


c

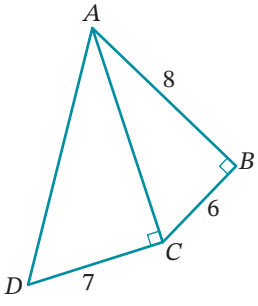


10 **a** Find the value of x, correct to 1 decimal place. R

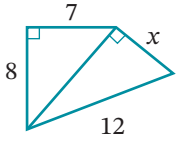
EXAMPLE 13



b Find the length of AD, correct to 2 decimal places. R



c Calculate the length of x as a surd. R



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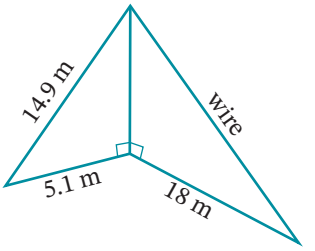
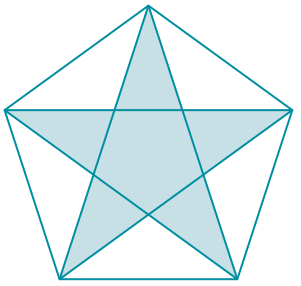
11 The diagonal of a rectangle is 200 mm in length. One of the side lengths of the rectangle is 167 mm. Find the length of the other side, correct to the nearest millimetre. P

12 To keep a power pole vertically upright, it is supported by 2 wires as shown. Find, correct to one decimal place, the length of the second wire. R



Did you know?

The Pythagoreans



The Pythagoreans were a group of men who were Pythagoras' followers. Apparently, they were so upset about the discovery of surds that they tried to keep it a secret. Hippasus, one of the Pythagoreans, was drowned for revealing the secret to outsiders.

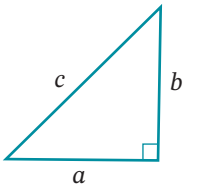
Find the special name for this symbol involving a star inside a pentagon, the symbol of the Pythagoreans.

2.06 Testing for right-angled triangles

Worksheet
Testing for right-angled triangles

Pythagoras' theorem says that if a right-angled triangle has sides of length a, b, and c, then $c^2 = a^2 + b^2$.

The reverse of this is also true:



The converse of Pythagoras' 'theorem'

If any triangle has sides of length a, b, and c that follow the formula $c^2 = a^2 + b^2$, then the triangle must be right-angled.

The right angle is the angle that is opposite the hypotenuse.

This is called the **converse** (or opposite) of Pythagoras' theorem, because it is the 'back-to-front' version of the theorem.

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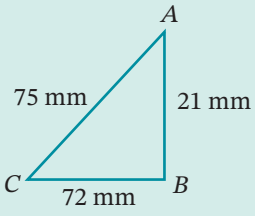
Example 14

Test whether $\triangle ABC$ is right-angled.



Solution

$75^2 = 5625$	Square the longest side
$21^2 + 72^2 = 5625$	Square the 2 shorter sides, then add
$\therefore 75^2 = 21^2 + 72^2$	The sides of this triangle follow $c^2 = a^2 + b^2$
$\therefore \triangle ABC$ is right-angled.	The right angle is $\angle B$

Example 15

Rahul constructed a triangle with sides of length 37 cm, 12 cm and 40 cm. Show that these measurements do not form a right-angled triangle.

Solution

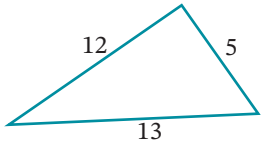
$40^2 = 1600$	Square the longest side
$12^2 + 37^2 = 1513 \neq 1600$	Square the 2 shorter sides, then add
$\therefore 40^2 \neq 12^2 + 37^2$	The sides of this triangle do not follow $c^2 = a^2 + b^2$
$\therefore$ The triangle is not right-angled.	

EXERCISE 2.06 ANSWERS ON P. 566

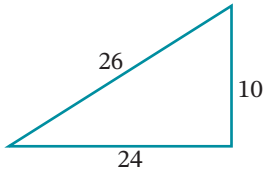
Testing for right-angled triangles U F R C

1 Test whether each triangle is right-angled. C

a



b



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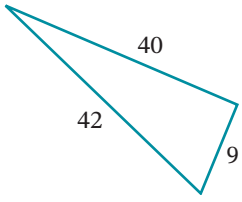

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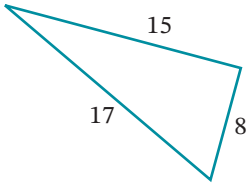
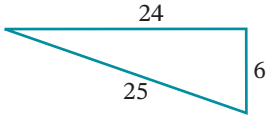
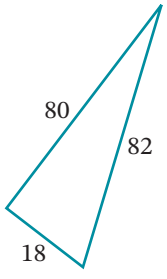


e



f



Video
Testing for right-angled triangles

2.06

EXAMPLE 14

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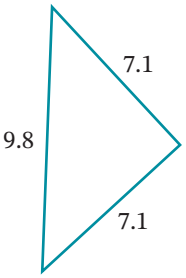
g



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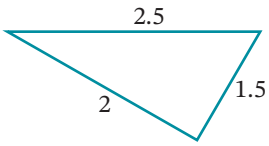
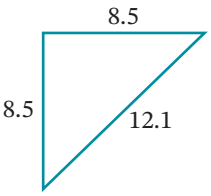


i



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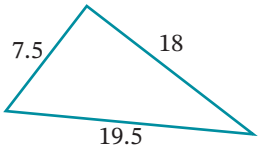
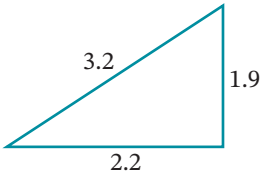
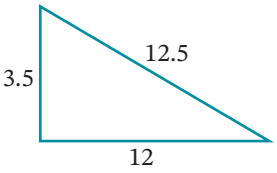


k



l



EXAMPLE 15

2 Paba constructed a triangle with sides of length 17 cm, 21 cm and 30 cm. Show that these measurements do *not* form a right-angled triangle. C

3 Which set of measurements would make a right-angled triangle? Select the correct answer **A**, **B**, **C** or **D**. C

A 2 cm, 3 cm, 4 cm

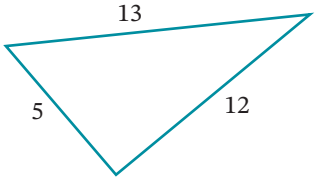
B 5 mm, 10 mm, 15 mm

C 12 cm, 16 cm, 20 cm

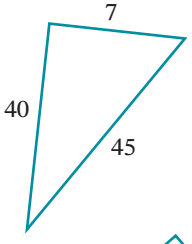
D 7 m, 24 m, 31 m

4 Which one of these triangles is not right-angled? Select **A**, **B**, **C** or **D**. C

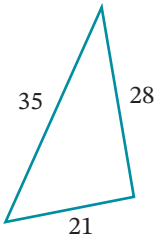
A



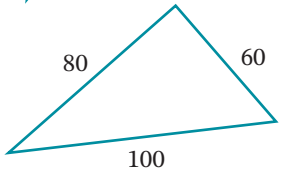
B



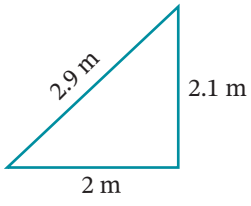
C



D



5 A pole 2.1 m tall is supported by a wire 2.9 m long. The other end of the wire is attached to the ground 2 m from the base of the pole. Is the pole standing vertically upright? R C



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Pythagorean triads

2.07

A **Pythagorean triad** or **Pythagorean triple** is any set of 3 positive integers that follow Pythagoras' theorem, for example, {3, 4, 5} or {5, 12, 13}. The word **triad** means a group of 3 related items ('tri-' means 3).

Worksheet
Pythagorean triads

Technology
Pythagorean triples

Spreadsheet
Pythagorean triples

Pythagorean triad

$\{a, b, c\}$ is a Pythagorean triad if $c^2 = a^2 + b^2$.

Any multiple of $\{a, b, c\}$ is also a Pythagorean triad.

Example 16

Test whether {5, 12, 13} is a Pythagorean triad.

Solution

$13^2 = 169$ — Squaring the largest number

$5^2 + 12^2 = 169$ — Squaring the 2 smaller numbers and adding them

$\therefore 13^2 = 5^2 + 12^2$ — These 3 numbers follow Pythagoras' theorem

$\therefore \{5, 12, 13\}$ is a Pythagorean triad.

Example 17

{3, 4, 5} is a Pythagorean triad. Create other Pythagorean triads by multiplying (3, 4, 5) by:

a 2 **b** 9 **c** $\frac{1}{2}$

Solution

a $2 \times \{3, 4, 5\} = \{6, 8, 10\}$

Checking: $10^2 = 100$
$6^2 + 8^2 = 100$
$\therefore 10^2 = 6^2 + 8^2$
$\therefore \{6, 8, 10\}$ is a Pythagorean triad.

b $9 \times \{3, 4, 5\} = \{27, 36, 45\}$

Checking: $45^2 = 2025$
$27^2 + 36^2 = 2025$
$\therefore 45^2 = 27^2 + 36^2$
$\therefore \{27, 36, 45\}$ is a Pythagorean triad.

c $\frac{1}{2} \times \{3, 4, 5\} = \{1.5, 2, 2.5\}$

Checking: $2.5^2 = 6.25$
$1.5^2 + 2^2 = 6.25$
$\therefore 2.5^2 = 1.5^2 + 2^2$
$\therefore \{1.5, 2, 2.5\}$ is a Pythagorean triad.

Strictly speaking, these are not Pythagorean triads because they are not whole numbers.

EXERCISE 2.07 ANSWERS ON P. 566

Pythagorean triads U F R C

EXAMPLE 16

1 Test whether each triad is a Pythagorean triad.

a {8, 15, 17}
b {10, 24, 26}
c {30, 40, 50}
d {5, 7, 9}
e {9, 40, 41}
f {4, 5, 9}
g {11, 60, 61}
h {7, 24, 25}
i {15, 114, 115}

2 Which of the following is a Pythagorean triad? Select the correct answer **A**, **B**, **C** or **D**.

A {4, 6, 8} **B** {5, 10, 12} **C** {6, 7, 10} **D** {20, 48, 52}

3 Use the spreadsheet you created in **Technology: Finding the hypotenuse** on page 57 to check your answers to questions **1** and **2**.

EXAMPLE 17

4 For each Pythagorean triad, create another Pythagorean triad by multiplying each number in the triad by:

i a whole number **ii** a fraction **iii** a decimal.

a {5, 12, 13} **b** {8, 15, 17} **c** {30, 40, 50} **d** {7, 24, 25}

Check that each answer follows Pythagoras' theorem.

5 Pythagoras developed formulas for finding Pythagorean triads $\{a, b, c\}$. If one number in the triad is a, the formulas for the other 2 numbers are $b = \frac{1}{2}(a^2 - 1)$ and $c = \frac{1}{2}(a^2 + 1)$.

a If $a = 5$, use the formulas to find the values of b and c.

b Show that $\{a, b, c\}$ is a Pythagorean triad.

6 Use the formulas to find Pythagorean triads for each value of a.

a $a = 7$
b $a = 11$
c $a = 15$
d $a = 4$
e $a = 9$
f $a = 19$
g $a = 10$
h $a = 51$

7 Prove that if $b = \frac{1}{2}(a^2 - 1)$ and $c = \frac{1}{2}(a^2 + 1)$, then $c^2 = a^2 + b^2$ is true. R C

8 There are many other formulas for creating Pythagorean triads. Use the Internet to research some of them.

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Pythagoras' theorem problems

2.08

When using Pythagoras' theorem to solve problems, it is useful to follow these steps.

- Read the problem carefully
- Draw a diagram involving a right-angled triangle and label any given information
- Choose a variable to represent the length or distance you want to find
- Use Pythagoras' theorem to find the value of the variable
- Answer the question

Worksheet
Applications of Pythagoras' theorem

Puzzle
Pythagorean two-step problems

Video
Pythagoras' theorem problems

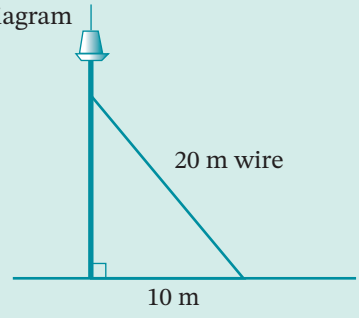
Example 18

A tower is supported by a wire that is 20 m long and attached to the ground 10 m from the base of the tower. How far up the tower does the wire reach? Answer to the nearest 0.1 m.

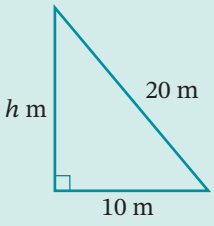
Solution

First draw a diagram. Let h represent how far up the tower the wire reaches in metres.

The diagram



The triangle



$20^2 = h^2 + 10^2$

$400 = h^2 + 100$

$h^2 + 100 = 400$

$h^2 = 400 - 100$

$= 300$

$h = \sqrt{300}$

$= 17.3205...$

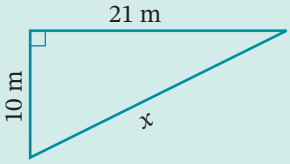
≈ 17.3

$\therefore$ The wire reaches 17.3 m up the tower.

Video
Pythagoras' theorem problems

Example 19

Find the perimeter of this triangle, correct to one decimal place.



Solution

Let x be the length of the hypotenuse.

$x^2 = 10^2 + 21^2$

$= 541$

$x = \sqrt{541}$

$\approx 23.3\text{m}$

Perimeter $\approx 10 + 21 + 23.3$

$= 54.3$ m

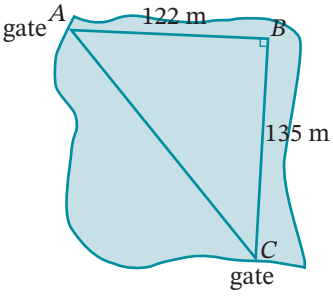
EXERCISE 2.08 ANSWERS ON P. 566

Pythagoras' theorem problems U F PS R

EXAMPLE 18

1 A flagpole is supported by a piece of wire that reaches 4 m up the flagpole. The wire is attached to the ground 2.4 m from the base of the flagpole. Find the length of the wire, correct to one decimal place. PS R

2 Mia is walking home from the library. She can walk around Hamper Park, entering the park at gate A, through B (stopping at the playground) and then out of the park at gate C, to her home nearby. Alternatively, Mia can take a shortcut across the park from gate A to gate C.



a Find the total distance Mia would walk from A to C via B.

b How far is it if Mia walks from A directly to C? Answer to the nearest metre.

c How much further would Mia walk if she takes the longer route home?

3 Cooper wanted to find the length XY of the lake shown. He placed a marker at Z so that $\angle YXZ = 90°$. He measured XZ to be 450 m long and ZY to be 780 m long. What is the length of the lake? Answer correct to one decimal place.

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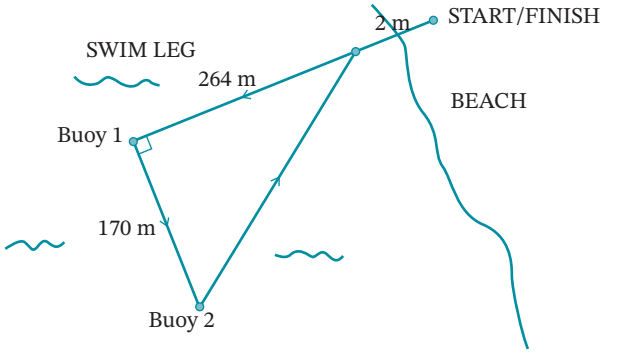
4 The swim course of a triathlon race has the shape of a right-angled triangle joined by a 2-metre link to the beach as shown. Calculate the total distance covered in this course (starting and finishing on the beach). PS

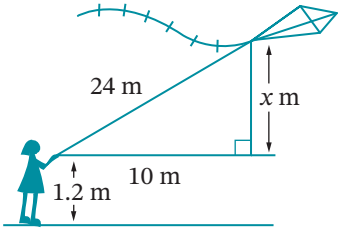


5 A kite is attached to a 24 m piece of string. The rope is held 1.2 m above the ground and covers a horizontal distance of 10 m.

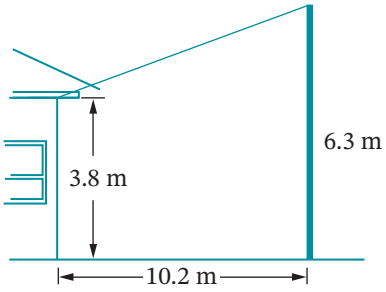
Find:

a the value of x, correct to one decimal place

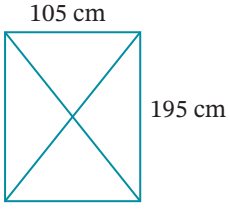
b the height of the kite above the ground, correct to the nearest metre.



6 Find the length of the electricity line from the top of the pole to the house, correct to one decimal place.



7 A screen door is reinforced by 2 metal braces that diagonally cross the door. Find the total length of the 2 braces in metres, correct to 2 decimal places.



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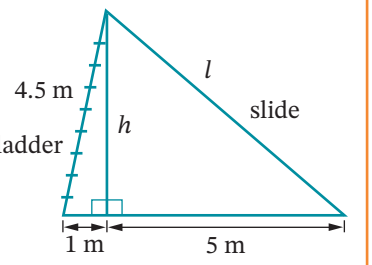
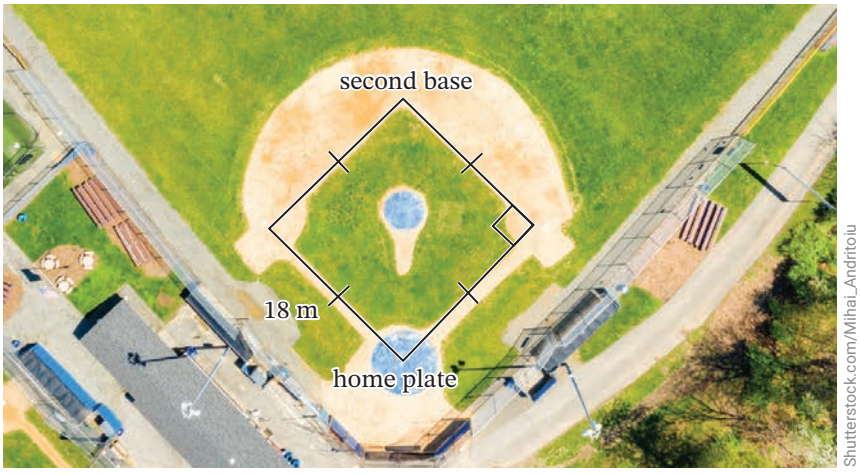
8 For this playground slide, find correct to 3 decimal places:

a the height (h) of the slide

b the length (l) of the slide.



9 A softball diamond has the shape of a square with a side length of 18 m. Sonja hits a ball from the home plate to second base. Calculate the distance travelled by the ball, correct to one decimal place. PS R



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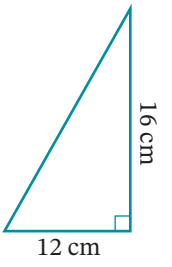
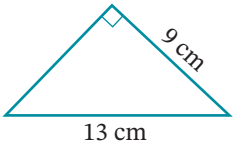
EXAMPLE 19

10 Calculate the perimeter of each shape, correct to one decimal place where necessary.

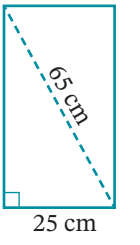
a



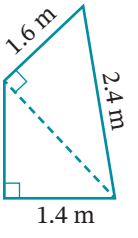
b



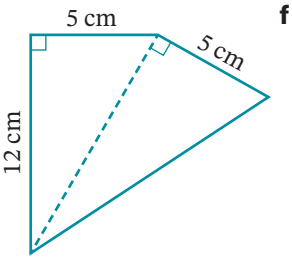
c



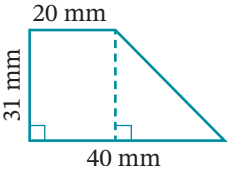
d



e



f



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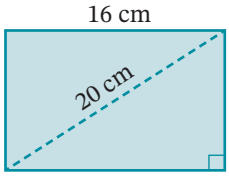
11 Calculate the area of each shape, correct to one decimal place where necessary.

a



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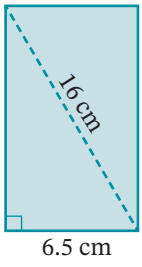

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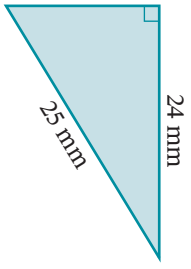
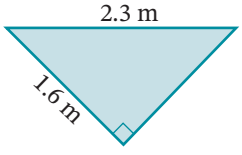
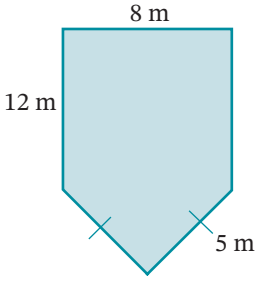


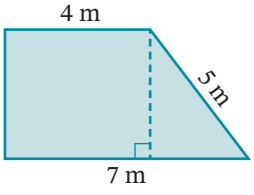
d



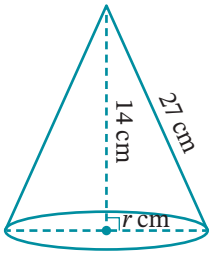
e



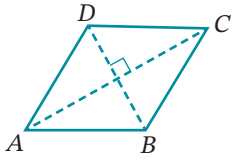
f



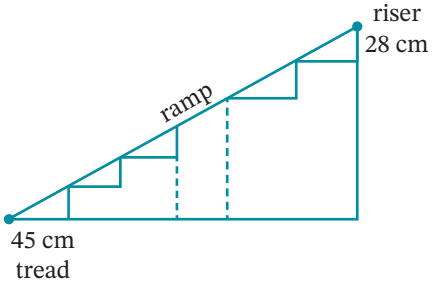
12 A cone has a vertical height of 14 cm and a slant height of 27 cm. Find r cm, the radius of its circular base, correct to 2 decimal places.



13 $DCBA$ is a rhombus with diagonals $AC = 18$ cm and $DB = 12$ cm. Find the perimeter of the rhombus, correct to one decimal place. PS R



14 A set of 24 steps is to be replaced by a ramp as shown. Each step has a tread of 45 cm and a riser of 28 cm. Find the length of the ramp in metres. PS R



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Quiz
Mental skills 2B

Mental skills 2B: Maths without calculators ANSWERS ON P. 566

Multiplying and dividing by a multiple of 10

1 Consider each example.

a $4 \times 700 = 4 \times 7 \times 100 = 28 \times 100 = 2800$

b $5 \times 60 = 5 \times 6 \times 10 = 30 \times 10 = 300$

c $12 \times 40 = 12 \times 4 \times 10 = 48 \times 10 = 480$

d $3.2 \times 30 = 3.2 \times 3 \times 10 = 9.6 \times 10 = 96$ (by estimation, $3 \times 30 = 90 \approx 96$)

e $4.6 \times 50 = 4.6 \times 5 \times 10 = 23 \times 10 = 230$ (by estimation, $5 \times 50 = 250 \approx 230$)

f $9.4 \times 200 = 9.4 \times 2 \times 100 = 18.8 \times 100 = 1880$ (by estimation, $9 \times 200 = 1800 \approx 1880$)

2 Now evaluate each product.

a	8×2000	**b**	3×70	**c**	11×900	**d**	2×300
e	4×4000	**f**	5×80	**g**	7×70	**h**	1.3×40
i	2.5×600	**j**	5.8×200	**k**	3.6×50	**l**	4.4×3000

3 Consider each example.

a $8000 \div 400 = 8000 \div 100 \div 4 = 80 \div 4 = 20$

b $200 \div 50 = 200 \div 10 \div 5 = 20 \div 5 = 4$

c $6000 \div 20 = 6000 \div 10 \div 2 = 600 \div 2 = 300$

d $282 \div 30 = 282 \div 10 \div 3 = 28.2 \div 3 = 9.4$

e $3520 \div 40 = 3520 \div 10 \div 4 = 352 \div 4 = 88$

f $8940 \div 200 = 8940 \div 100 \div 2 = 89.4 \div 2 = 44.7$

4 Now evaluate each quotient.

a	$560 \div 70$	**b**	$2500 \div 500$	**c**	$3200 \div 400$	**d**	$440 \div 20$
e	$160 \div 40$	**f**	$1500 \div 30$	**g**	$450 \div 50$	**h**	$744 \div 80$
i	$2550 \div 300$	**j**	$846 \div 200$	**k**	$576 \div 60$	**l**	$2160 \div 90$

Did you know?

Pythagoras and President Garfield

According to *The Guinness Book of Records*, Pythagoras' theorem has been proved the greatest number of times, with a book published in 1940 containing 370 different proofs, including one by US president James Garfield (1831–1881). Before becoming involved in politics, Garfield studied mathematics and taught in a public school. He published his proof in 1876 and was the 20th president of the United States, governing for only 150 days in 1881.

Find the main mathematical feature of President Garfield's proof.

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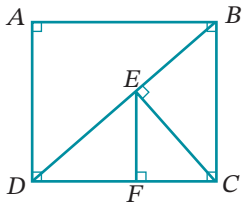
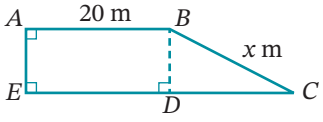
Power plus ANSWERS ON P. 566

1 Name the 5 hypotenuses on this diagram.

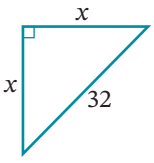


2 The rectangle $ABDE$ in the diagram has an area of 160 m^2. Find the value of x as a surd if $\triangle BDC$ is an isosceles triangle.

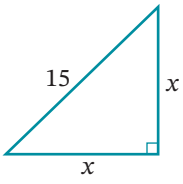


3 Find the value of x in each triangle, correct to one decimal place.

a



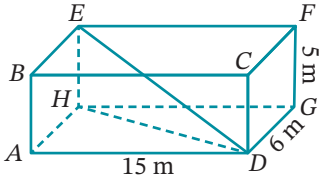
b



4 For this rectangular prism, find, correct to one decimal place, the length of diagonal:

a HD

b DE

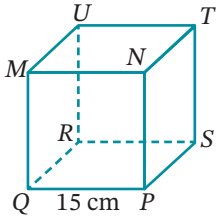


5 Find, correct to one decimal place, the length of the interval joining points $A(-2, 1)$ and $B(3, -2)$ on the number plane.

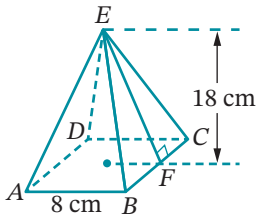
6 For this cube, find, correct to 2 decimal places, the length of diagonal:

a QS

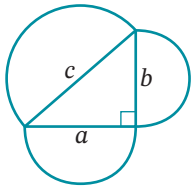
b QT



7 For this square pyramid, find, as a surd, the slant height EF.



8 Use Pythagoras' theorem to prove that for a right-angled triangle, the area of the semicircle constructed on the hypotenuse is equal to the sum of the areas of the semicircles constructed on the other 2 sides.



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CHAPTER 2 REVIEW

Language of maths

area	converse	diagonal	formula
hypotenuse	irrational	perimeter	Pythagoras
right-angled	shorter	side	square root
surd	theorem	triad	unknown

1 Who was **Pythagoras** and which country did he come from?

2 Describe the **hypotenuse** of a right-angled triangle in 2 ways.

3 What is another word for 'theorem'?

4 For what type of triangle is **Pythagoras' theorem** used?

5 What is a **surd**?

6 What is the name given to a set of 3 numbers that follows Pythagoras' theorem?

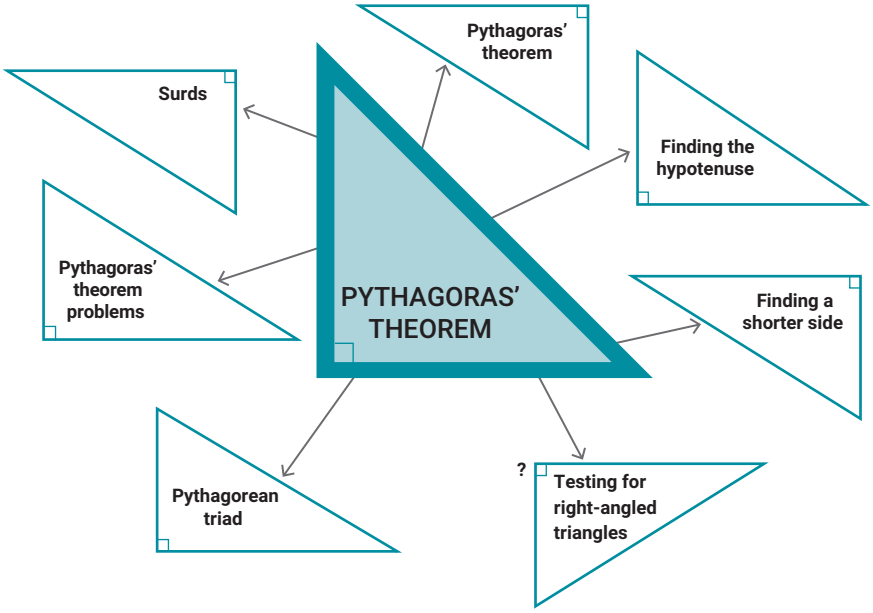
Quiz
Language of maths 2

Puzzle
Pythagoras' theorem find-a-word

Topic summary

- How relevant do you think Pythagoras' theorem is to our world? Give reasons for your answer.
- Give 3 examples of jobs where Pythagoras' theorem would be used.
- What did you find especially interesting about this topic?
- Is there any section of this topic that you found difficult? Discuss any problems with your teacher or a friend.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Worksheet
Mind map: Pythagoras' theorem

Quiz
Test yourself 2

TEST YOURSELF 2

ANSWERS ON P. 566

2.01 **1** Evaluate each expression.

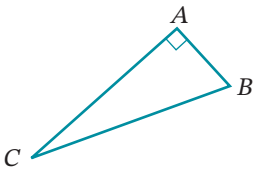
a 8.5^2 **b** $(-6)^2$ **c** $9^2 + 11^2$

2.01 **2** Evaluate each expression correct to 2 decimal places.

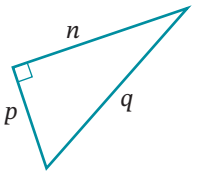
a $\sqrt{146}$ **b** $\sqrt{6^2 + 12^2}$ **c** $\sqrt{27^2 - 16^2}$

2.02 **3** Name the hypotenuse in each triangle.

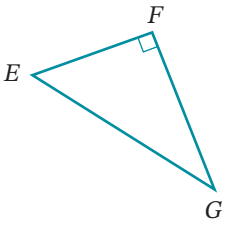
a



b



c

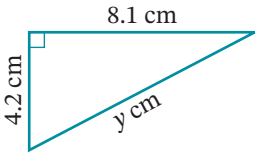


2.02 **4** Write Pythagoras' theorem for each triangle in question **3**.

2.03 **5** Find the value of y. Select the closest answer **A**, **B**, **C** or **D**.

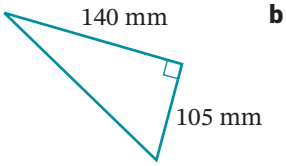
A 12.3 **B** 9.1

C 6.9 **D** 3.5



2.03 **6** For each triangle, find the length of the hypotenuse. Give your answer as an exact value.

a



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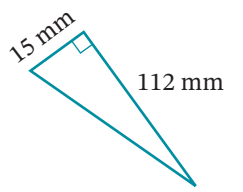
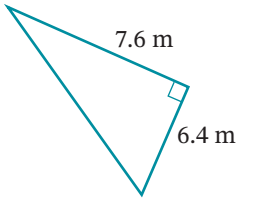


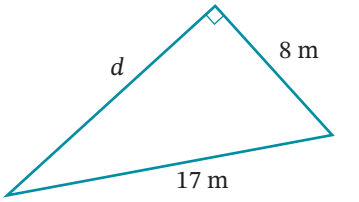
c



2.04 **7** Find the value of d. Select **A**, **B**, **C** or **D**.

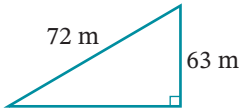
A 12 m **B** 15 m

C 16.8 m **D** 18.9 m

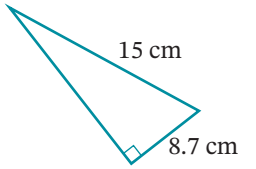


2.04 **8** For each triangle, find the length of the unknown side. Give your answer correct to one decimal place.

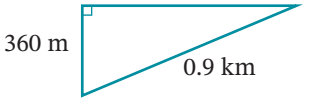
a



b



c



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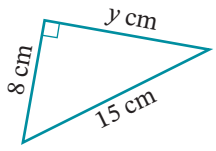
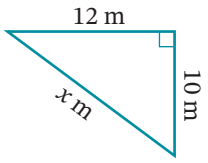
9 Find the exact value of the variable in each triangle. 2.05

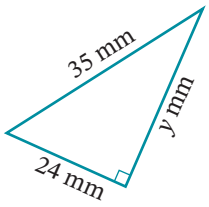
a



b



c



10 In $\triangle DEF$, $\angle F = 90°$, $DF = 84$ cm, and $EF = 1.45$ m. Find DE, correct to the nearest centimetre. 2.05

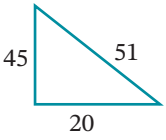
11 Julie walks 6 km due east from a starting point P, while Luka walks 4 km due south from P. 2.05

a Draw a diagram showing this information.

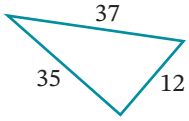
b How far are Julie and Luka apart? Give your answer correct to one decimal place.

12 Test whether each triangle is right-angled. 2.06

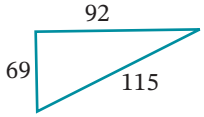
a



b



c



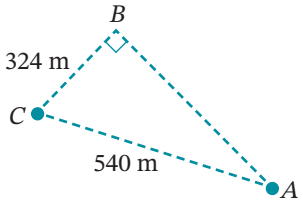
13 Test whether each triad is Pythagorean. 2.07

a {7, 24, 25} **b** {5, 7, 10} **c** {20, 21, 29} **d** {1.1, 6, 6.1}

14 On a cross-country ski course, checkpoints C and A are 540 m apart and checkpoints C and B are 324 m apart. Find the distance: 2.08

a between checkpoints A and B

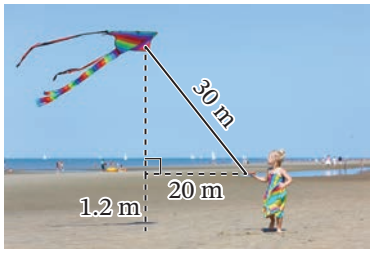
b skied in one lap of the course in kilometres.



15 Miranda holds a kite string 1.2 m above the ground. How high is the kite above the ground? Select **A**, **B**, **C** or **D**. 2.07

A 22.4 m **B** 23.6 m

C 36.1 m **D** 37.3 m



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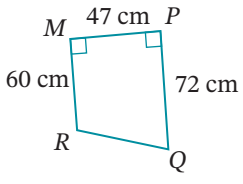
16 Find the length of QR in this trapezium, correct to one decimal place. 2.08



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